Office Medium Prototype Scorecard Spreadsheet Review

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December 9, 2025

On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. Changes shown in square brackets [].

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- IT Rooms hould have people based on Title 24-2025 Table 120.1-A (OFFM26-Zones/OFFM38-HVAC System)
- SHGC for 1998-2007 wrong value for several non-north CZs (OFFM31-Envelope)
- Infiltration for 2008-2018 slightly off (OFFM36-Envelope)
- Wrong EER used for VRF (OFFM40-HVAC System)
- Total static pressure for existing vintages incorrect (OFFM42-HVAC System)
- Top floor breakroom should have hot water usage (OFFM49-Water Heater)
- Restroom lighting power density too low (OFF53-Interior Lights)
- Lobby lighting power density too low (OFF56-Interior Lights)
- Plug load for some spaces incorrect (OFFM61, OFFM62)
- Service hot water schedule incorrect (OFF67-Schedule)

The following are the highest priority comments:

- None

The following comments represent other important issues found:

- Use punched windows instead of curtain walls (OFFM13-Prototype)
- Ground floor lobby space cannot be in core (OFFM16-Zones)
- Mechanical rooms should not be cooled (OFFM18-Zones)
- IT Rooms should not share HVAC system with other core spaces (OFFM19-Zones)
- Middle and top floor “lobbies” should have higher occupancy (OFFM24-Zones)
- Private offices too small (OFFM29-Zones)

- Use calculated capacity to determine cooling equipment size category (OFF39-HVAC System)
- IT Rooms, stairways and storage should have their own thermostat scheduls (OFF44-Thermostat)
- Several spaces should have electric water heaters (OFFM48-Water Heater)
- Additional lighting power typically used for offices and lobbies (OFFM58-Interior Lights)
- IT Room plug load power density should be justified (OFFM63-Equipment)
- Saturday schedules not well justified (OFFM65-Schedules)
- Schedules for lobbies and breakrooms should not be same as office occupants (OFFM69-Schedules)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Heating efficiency not explained (OFFM41-HVAC) [added to this list]
- No explanation on why 125 gal/hr is used for water heater (OFFM50-Water Heater) [added to this list]
- Why is the ambient temperature for the water heater assumed to be 68F (OFFM51-Water Heater) [added to this list]

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Ext Lighting
- Equipment

- Schedules
- Weights
- Energy Usage
- PV
- EV Charger
- Data

The comments on these tabs are labeled using the following format OFFMx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

Geometry of CalBEM Office Prototypes Memorandum

As part of reviewing the scorecard spreadsheet, the November 13 version of the "Geometry of CalBEM Office Prototypes Memorandum" was reviewed. Formatting issues found were:

- In Figure 1, the graphs for the heating system and cooling system seem to be missing the meaning of the yellow color, which is the majority of the systems for smaller buildings.
- The report shows four tables shown as "Table 4"
- Table 8 appears before Table 5
- Table 12 appears without Tables 9 to 11

The technical analysis seems sound.

OFFM1-Notes (n/a)

The information from the "Geometry of CalBEM Office prototypes" memorandum should be summarized and put on the Notes tab.

OFFM2-Notes (n/a)

Rather than repeat it, please see comment PPD1 in my single-family residential prototype review, LHSC1-Notes in the large hotel review, SHSC1-Notes in the small hotel review, and HOSP1-Notes in the hospital review.

OFFM3-Notes (n/a)

Rather than repeat it, please see comment PPD2 in my single-family residential prototype review, LHSC2-Notes in the large hotel review, SHSC2-Notes in the small hotel review, and HOSP1-Notes in the hospital review.

OFFM4-Notes (n/a)

Many references related to the characteristics of office buildings exist covering statistics and typical practice since office buildings are one of the most common type of commercial buildings. I did not perform a literature search for this building type since it was familiar to me but the usual source such as the national labs, especially PNNL and NREL, and ASHRAE should be examined. Two space planning references that may be useful that are on my shelf which may not be found during a literature search for office buildings are:

- Space Planning Basics – Mark Kelen and Rob Fleming
- Space Planning for Commercial Office Interiors – Mary Lou Bakker

A more thorough literature search for statistics and best practices would be warranted especially given that three office building prototypes are being used.

OFFM5-Notes (n/a)

How do the assumptions for this model match up to the DEER model, to the other office prototypes, to the PNNL prototype, to the CEC medium office from CBECC, to the ORNL medium office? It seems like a comparison table would be especially useful, showing the area and fraction of area for each space type.

OFFM6-Notes (A35:H39)

Currently, a single version of Title 24 is chosen to be typical for construction that occurs over many different versions of Title 24. For example, Title 24-1986 is used for the vintage

that is intended to represent all construction from 1984 to 1997, which actually had six different updates to Title 24. An alternative to this approach is to attach the weight of construction from the Dodge data and use that to create a weighted average across all different Title 24 versions that occurred over the time period.

OFFM7-Notes (B2:C2, B10:C10, B13:C13)

Please add links for Dodge 2023, Title 24 part 6, ASHRAE 90.1 (2022), and ACM (2022).

OFFM8-Notes (B24:B33)

Please add the number as a table that corresponds to the values depicted in the graph for construction rates based on Dodge data.

OFFM9-Notes (n/a)

Office buildings vary based on the Class of the building. Class A buildings are expected to serve high-end markets and are likely to have greater amenities, including different finishes (including more decorative lighting) than Class B or C buildings, which are more intermediate or basic buildings. This is likely to be an important distinction in expectations of the building that will impact energy use. Please identify which office building class has been assumed for the prototype.

OFFM10-Notes (n/a)

Many of the abbreviations being used on the data sources are familiar to building energy analysts but may not be as familiar to others. Please add an unabbreviated version of each item.

OFFM11-Prototype (D6)

The link to the document "NonRes-Vintage bins-Draft Report-30Nov2023-Final.docx" does not seem to work.

OFFM12-Prototype (A12:C12)

Several references to sources are made to the window-to-wall ratio to justify the 33% value, including PNNL, CEC, CBECS, Comstock, and DEER. Given the historic adoption of the PNNL model by others, it is not too surprising that 33% is used for several of these. It would be good to go back to surveys and decide if the fraction is reasonable. I believe that if medium office buildings are defined as office buildings between 2 and 6 stories, the fraction may be too high due to many of these buildings not having as much glazing on the non-street side of the buildings in urban areas.

OFFM13-Prototype (A12:C12)

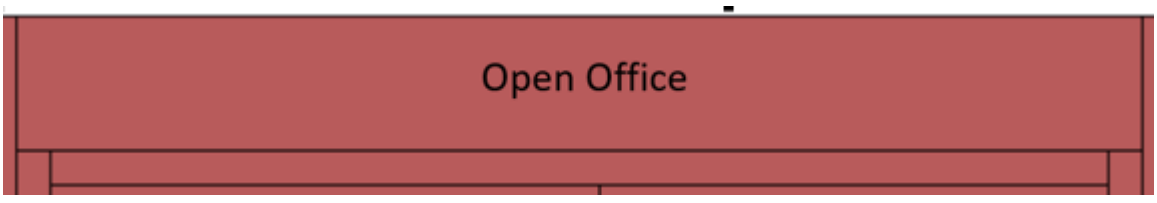
While very long windows stretching from corner to corner may be used in some high-rise office buildings with curtain walls or with spandrel glass panels to attain a specific aesthetic, they are much less common in low-rise office buildings, which are generally built as inexpensively as possible. Walls with punched windows are cheaper than curtain walls, and the location of the glass can be optimized to best serve the spaces within.

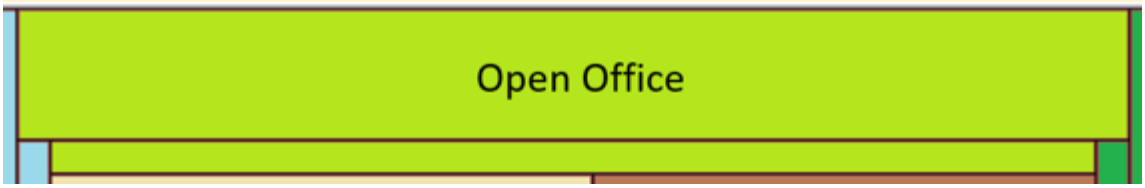
OFFM14-Zones (M2:S36)

Why is the zoning layout shown twice? The heading shows "Layout," and the top figure is labeled "Layout of Spaces," and the bottom figure is showing "Layout of Zones," but they are the same layout, other than colors. If the intention is that each different color represents a zone, then perhaps a key should be added for the color and the zone name. In addition, the table column A is described as Space/Thermal Zone, which implies that they are the same (one space per zone). It is just not very clear how the spaces belong to the zones, and the pair of "Layouts" does not help answer that question.

OFFM15-Zones (M2:S36)

Why do the exterior spaces in the layout diagram and the zones in the zoning diagram have a strange stairstep in their shapes? The Open Office shown is one example, but it applies to all perimeter spaces and zones in both diagrams.





What is the purpose of the second narrow rectangular part of the space/zone?

OFFM16-Zones (A9:B9, A27:B27, A45:B45)

The "Lobby/Lounge_FloorG", "Lobby/Lounge_FloorM", and "Lobby/Lounge_FloorT" are presumably the Lobby that appears in the Layout figures and is shown in the core of the building, without an exterior exposure. How can a "Lobby" be in a core zone and not adjacent to an exterior wall on the ground floor, since it would presumably have a door that people would enter from outside, especially since its "Space Type" is called "Lobby, Main Entry"? I guess a "Lounge" part that applies to the middle and top floors and could be completely interior, but that seems also unlikely since I would expect a lounge to have windows. How is a "lounge" different from a "break" room, which is also defined? Calling this a lobby on the second and third floors really doesn't make sense. The name should probably be updated on the figure and the table.

OFFM17-Zones (A10:E10, A28:E28, A46:E46)

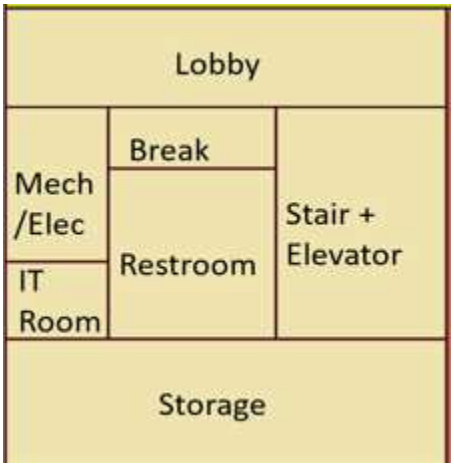
If "Storage_1" space has zero occupancy all the time, should it be conditioned? This also applies to the second and third floors.

OFFM18-Zones (A11:E11, A29:E29, A47:E47)

Mechanical/Electrical_FloorG, Mechanical/Electrical_FloorM, Mechanical/Electrical_FloorT have a space type of Electrical, Mechanical, Telephone Rooms, but are shown as conditioned. Typically, these spaces are unconditioned or semi-conditioned.

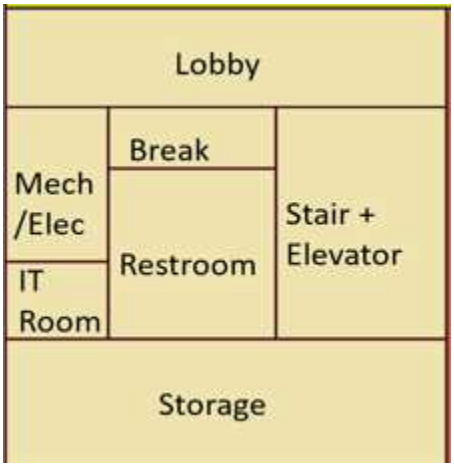
If they are unconditioned or semi-conditioned, presuming that the color indicates a zones, the core zone shown to the left on the figure, and shown below, should not contain both

conditioned and unconditioned or semi-conditioned spaces like the "Mech/Elec" space (Mechanical/Electrical_FloorG, Mechanical/Electrical_FloorM, Mechanical/Electrical_FloorT), a zone needs to be serviced by a single HVAC system that would not be able to do that.



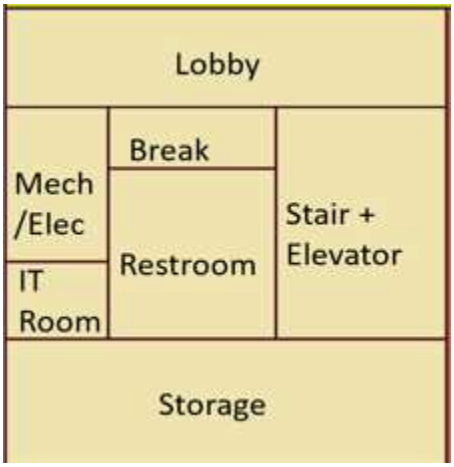
OFFM19-Zones (A12:E12, A30:E30, A48:E48) and on Zones-Equipment (A8:B8)

The core zone on the left side in the diagram shown below (presuming that a color indicates a zone) should not contain both spaces with normal interior loads and spaces with very high interior loads, like the "IT Room" (IT-Room_FloorG, IT-Room_FloorG, IT-Room_FloorG). The loads on the Equipment tab show it having 12 W/sqft compared to less than 1 W/sqft for most other spaces in that zone. This is a poor modeling practice.



OFFM20-Zones (A13:C13, A31:C31, A49:C49)

Presuming the colors in the diagram indicate the zoning, the core zone to the left, shown below, should not contain both spaces with typical ventilation requirements and those that require special ventilation/exhaust requirements like the Restroom_FloorG, Restroom_FloorM, and Restroom_FloorT spaces. A zone needs to be serviced by a single HVAC system that would not be able to do that.



OFFM21-Zones (A15:C15, A33:C33, A51:C51)

If "Stairway_1" space has zero occupancy all the time, should it be conditioned? For fire-escape type of stairs, they are often unconditioned. This also applies to the second and third floors.

OFFM22-Zones (A60:A62)

The bottom of the table on the Zone tab shows:

> Ref: Geometry and zones: ORNL 2019 medium office prototype with modification
>
> https://web.eecs.utk.edu/~jnew1/publications/2019_IBPSA_OpenStudioOffice.pdf
>
> https://web.eecs.utk.edu/~jnew1/presentations/2019_IBPSA_OpenStudioOffice.pdf

No justification is provided for the modifications. The ORNL medium office, shown below, has a more detailed layout that was based on data from actual buildings in PNNL's National Commercial Construction Characteristics NC3 database, based on Dodge data.

The original ORNL medium office shows

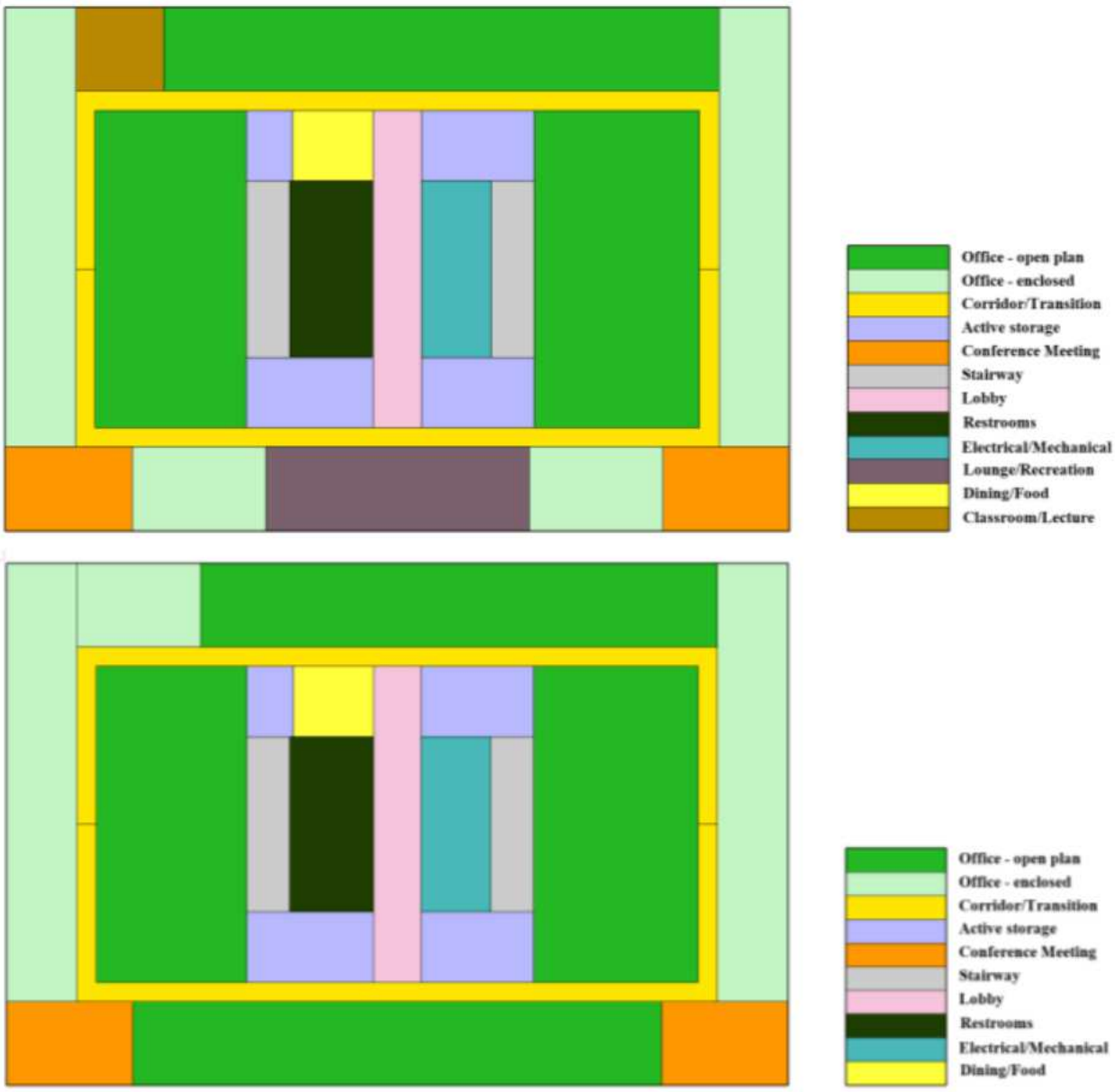
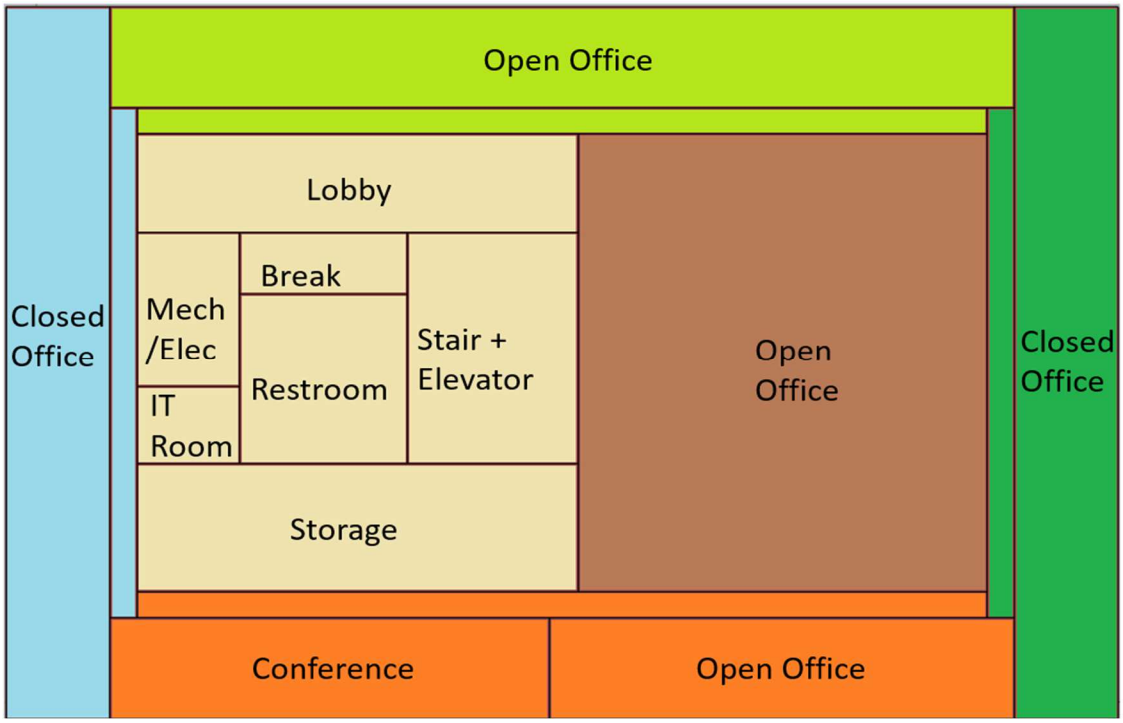


Figure 4. New medium office prototype model layout – ground floor (top), mid and top floors (bottom)



OFFM23-Zones (C2:C57)

Please populate the HVAC System Name/AHU Name. It is difficult to completely understand the zone and system assignment without this information. If it is different by vintage, perhaps add multiple columns.

OFFM24-Zones (K27:L27, K45:L45)

The value of 10 persons/1000ft2 from ASHRAE Standard 62.1 is based on the description from the standard "Main Entry Lobby," but this would not be appropriate for the middle and top floors of a building. Perhaps "Breakroom" at either 33 (T24) or 50 (62.1) persons/1000ft2 would be more appropriate.

OFFM25-Zones (N/A)

Please indicate the exact category from Title 24-2025 Table 120.1-A that is being used; many of the spaces are based on interpretation, and different people may interpret the space use differently.

OFFM26-Zones (K10, K28, K48)

The value of zero people/1000ft2 for an IT room is not consistent with the 5 people/1000ft2 shown in Title 24-2025 Table 120.1-A for "Computer (not printing)."

OFFM27-Zones (K12, K30, K46)

The value of zero people/1000ft2 for a storage room is not described in the Title 24-2025 Table 120.1-A, and the secondary source of ASHRAE 62.1-2019 (although I think more recent versions have the same value) shows 2 people/1000ft2 for "Occupiable storage rooms for dry materials."

OFFM28-Zones (n/a)

Please provide a table of area by space type. It is difficult to understand the overall composition of the building without such a breakdown. The office and conference room space takes 68% of the building, which seems reasonable.

OFFM29-Zones (G5:J6, G23:J24, G41:J45)

The average size of a closed office would be 205 ft2, based on 1638 ft2 and 8 people. This is small for private offices that are usually 225-300 ft2. The value of 205 ft2 is reasonable for open offices.

OFFM30-Envelope (H5:H7, H10:H12, H15:H17, H22:H23)

The Notes column does not describe the table the values were taken from, as they are described for other prototypes. This makes it more difficult to verify the values.

OFFM31-Envelope (H20:H21, J21, N21:W21)

The SHGC for the 1998-2007 vintage using Title-24-2001 shows the use of the window-to-wall ratio bin of 20-30% but the prototype is 33% so it should really use the 31-40% bin. Due to this, several non-North values are incorrect for CZ3 and CZ6 through CZ15.

OFFM32-Envelope (G9)

The roof values are based on Title 24-1992, as explained in the note: "Considering 30 years for roof/ insulation replacement. (Guest room requirement)" and the values are correct, but the presumption that T24-1992 is a good stand-in for this in 2025 seems incorrect. If one assumes that all these roofs were replaced 30 years ago, then Title 24-1995 would make more sense. Since the odds of all the roofs being replaced in 30 years are slight, a better assumption would be:

- The average value of roof requirements from Title 24 from 1995, 1998, 2001, 2005, 2008, 2010, 2013, 2016, 2019, 2022, or
- The value of roof requirements from 15 years ago, so using Title 24-2010 (less accurate but still justifiable).

This is a repeat of the comment for the large hotel LHSC26-Envelope.

OFFM33-Envelope (F13:X14, F18:X19)

Averaging the U-Factor and SHGC between single and double-pane options leads to non-physically possible windows. The weighting between single and double pane is based on a statistic that showed 50% of medium offices that were built before 2000 were single pane from CBECS 2018. The weighted value is shown correctly, but the concept of ending up with a type of fenestration that is not physically possible may cause problems with simulation engines such as EnergyPlus that expect a window to be built up from a specific pane of glass with specific properties. It might be better to have a mixture of single-pane and double-pane windows in a model (although that is a different compromise with reality). This is almost an exact repeat of the comment for the large hotel LHSC29-Envelope and small hotel SHSC39-Envelope.

OFFM34-Envelope (A1:X23)

The main envelope table was reviewed. All values were verified except those described in other comments.

OFFM35-Envelope (C3:E12)

The use of a wood-framed roof may be inconsistent with the use of steel-framed walls.

OFFM36-Envelope (E25:X31)

To verify the infiltration values, the four and six side areas were computed:

- Area 4 sides = (13 ft x 3 ft) x (163.7 + 109.2) x 2 = 39 x 545.8 = 21,286.2 ft2
- Area 6 sides = Area 4 sides + 163.7 x 109.2 x 2 = Area 4 sides + 35752.1 = 57,038.3 ft2
- Ratio of 6 sides to 4 sides = 57,038.3 /21,286.2 = 2.680

So, according to the referenced report

$$I_{design} \text{ at the building envelope } = 0.112 \times I_{75} \text{ at building envelope} \qquad 2$$

$$I_{design\ (4-side)} = I_{design\ (5-or\ 6-side)} \times \left(\frac{\text{Total leakage area } (5-\text{ or } 6-\text{ side})}{\text{Exterior Wall area}}\right) \qquad \textit{Equation 3}$$

Using these equations, the values should be:

Pre 1978	0.3302
1978-1997	0.3302
1998-2007	0.3302
2008-2018	0.3002
2019+	0.1801

The 2008-2018 values does not match the value shown in the infiltration table.

OFFM37-HVAC System (B3:B4)

Having a single HVAC system per zone for the core zone does not make sense for the room with IT equipment. This is very unlikely to be zoned that way. The high loads for each IT room would likely prompt an HVAC system to serve just those spaces.

OFFM38-HVAC System (M11:M12)

Typo in the size category. It says "65,000-13,5000," and it should be "65,000-135,000" with the comma moved between the third and fourth digits.

OFFM39-HVAC System (L10:L25)

Estimating the cooling capacity size by using 20 (Btu/hr)/sqft multiplied by the areas of the zone seems inexact for a simulation-based analysis. It would make more sense to use the capacity estimated by the simulation tool to determine the size category and thus the efficiency of the DX cooling equipment for each zone for each climate zone.

OFFM40-HVAC System (N23:N25, P23:P25, C15:D15, E15)

The cooling EER for new construction using Title 24-2025 and assuming VRF as described in the first table on the tab is a heat pump VRF (since it is shown for both heating and cooling), the correct table the EER values should be taken from is Table 110.2-G. There are many options, but presuming air-cooled and greater than 240,000 Btu/h, in cooling mode, the value is 9.1 EER, and in heating mode is 2.25 COP at 17db/15wb and the federal minimum COP at 47db/43wb (I think this is 3.3 COP). In addition, the footnote to Table 110.2-G says "Deduct 0.2 from the required EERs and IEERs for Variable Refrigerant Flow (VRF) Multi-split system units with a heating recovery section." So if heat recovery is used in the model, these values should be further reduced.

The value of 10.0 EER in the table is for standard packaged equipment, not VRF.

OFFM41-HVAC System (C10:C15, E10:E15)

The heating efficiency column is not explained well. The source is shown as Table 112-A, which is not right for heating equipment. The 80% efficiency value seems to be for furnaces, but it is not explained. The COP=3 for the 2008-2018 vintage implies the use of a heat pump, but that is not assumed in the table immediately above for existing equipment. Likewise, the 80% efficiency for new construction implies the use of a furnace, but in the table above, VRF is used for heating. Also see the previous comment, OFFM40 -HVAC System, about the proper values to show in this summary table based on VRF.

OFFM42-HVAC System (D31:D32)

The total static pressure (TSP) shown in the table for CAV>7437 cfm and VAV>4648 cfm is incorrect for existing vintages. The table these are based on the PNNL report has another size category between those cfm's and 20,000 cfm, see below. The values in the scorecard are for the >20,000 cfm values. All three values should probably be reflected in the scorecard table, or the description should be changed to ">20,000 cfm."

Table 5.11. Rules for Establishing Fan Power Inputs for PI Analysis

System	CFM Range	TSP in. w.c.	Fan Efficiency
Fan Power Limitation Rule for 90.1-2004			
CAV	< 7,437	2.50	0.65
VAV	< 4,648	4.00	0.65
CAV	≥ 7,437 and < 20,000	4.46	0.65
CAV	≥ 20,000	4.09	0.65
VAV	≥ 4,648 and < 20,000	6.32	0.65
VAV	≥ 20,000	5.58	0.65
Unit Heater	All	0.2	0.65
PTAC, PTHP, or FCUs	All	1.33	0.65
Fan Power Limitation Rule for Addendum 90.1-04ac			
CAV	< 7,437	2.50	0.65
VAV	< 4,648	4.00	0.65
CAV	≥ 7,437	4.09	0.65
VAV	≥ 4,648	5.58	0.65
Unit Heater	All	0.2	0.65
PTAC, PTHP, or FCUs	All	1.33	0.65

OFFM43-HVAC System (F35)

For new construction, a fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction. This is the same comment as LHSC52-HVAC System.

OFFM44-Thermostat (n/a)

The values shown on the Thermostat tab of the scorecard spreadsheet are consistent with the source of Title 24-2022 Appendix 5.4B. A few of the zones could potentially have special thermostat schedules, such as:

- IT-Room – it is unlikely to be allowed to go to 85 at night during setback
- Stairway – it may not be cooled at all, and the heating setpoint may be lower than 70°F during the day and may not have setback.

- Storage – it may not be cooled at all, and the heating setpoint may be lower than 70°F during the day and may not have setback.

OFFM45-Ventilation (N/A)

Please indicate the exact category from Title 24-2025 Table 120.1-A that is being used; many of the spaces are based on interpretation, and different people may interpret the space use differently.

OFFM46-Ventilation (n/a)

No differentiation by vintage was included for outdoor airflow. Earlier vintages of offices probably had significantly different ventilation requirements. Since this often impacts the sizing of ducts, intake vents, and air-handling equipment, it is unlikely to be upgraded very often. This is the same comment as LHSC58-Ventilation.

OFFM47-Water Heater (A4:B4)

For new construction, using a gas water heater is unclear based on the language of the ACM for water heater type and size:

> Standard Design: For healthcare facility spaces, the same as proposed. For all other spaces, the standard design shall have one gas storage water heater if any of the spaces have a space water heating fuel type of gas (from Appendix 5.4A), and the standard design building will have one electric water heater if the any of the spaces have a space water heating fuel type of electric.

For buildings with a mixture of gas and electric spaces, the description above seems unclear.

OFFM48-Water Heater (B9:B63)

The Baseline WH Type column shows either "Gas" or is blank for all values in the spreadsheet, but when looking at the cited source Title-24 2022, Appendix 5.4A, sheet: "2022 SpaceFuncData-Input", the values should be "Electric" for the following space types:

- Computer Room
- Convention, Conference, Multipurpose and Meeting Area
- Corridor Area
- Electrical, Mechanical, Telephone Rooms
- Office Area (>250 square feet)
- Office Area (<250 square feet)
- Stairwell
- Storage, General

OFFM49-Water Heater (E57)

The breakroom "Break_FloorT" is shown with no hot water usage specification, but should be 0.09 gal/hr-person, as the breakrooms on the other floors are.

OFFM50-Water Heater (B70)

No explanation is given for the formula and why 125 gal/hr is used instead of the sum of the actual estimated usage values.

OFFM51-Water Heater (B81)

No source is provided for the assumption of using 68°F for the ambient temperature.

OFFM52-Water Heater (B75)

The cold water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater.

OFFM53-Interior Lights (B10)

The restroom lighting power density for new construction is shown as 0.6 W/ft2, but Table 140.6C shows 0.65 W/ft2.

OFFM54-Interior Lights (B9)

The IT Room lighting power density for new construction is shown as 1.1 W/ft2, but the cited Table 140.6C does not show a corresponding value. Please explain how this was derived.

OFFM55-Interior Lights (G4:G14)

Using Title-24 2013 for all existing buildings may not be as accurate as taking an average of multiple versions of Title 24 covering recent vintages, assuming that older buildings have replaced their lighting in the last 20 years (or some other time period).

OFFM56-Interior Lights (E6)

The existing LPD for Lobby, Main Entry is shown as 1.1 W/ft2, but it is 1.5 W/ft2 in the cited source of Title 24-2013 Table 140.5C for "Lobby Area – Main Entry Lobby."

OFFM57-Interior Lights (E9)

The IT Room lighting power density for existing buildings is shown as 1.2 W/ft2, but the cited Title 24-2013 Table 140.6C does not show a corresponding value. Please explain how this was derived.

OFFM58-Interior Lights (n/a)

The following space types allow additional lighting power, and it is very likely that for many of these space types, the additional lighting power would be used:

- Office Area (>250 square feet)
- Lobby, Main Entry

OFFM59-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72-Ext Lights and SHSC67-Ext Lights.

OFFM60-Ext Lights (B6)

Please document the sources used for assumptions for computed areas and counts. It mentions the 2019 Outdoor Lighting Case Report and PNNL 90.1-2010 TSD, but those are both big, complicated documents. More specific references would enable the values to be checked.

OFFM61-Equipment (B4)

The plug load power density of 0.8 W/ft2 for "Convention, Conference, Multipurpose and Meeting Area" is not consistent with the 1.0 W/ft2 that appears in the source T24 2022 - Appendix 5.4A.

OFFM62-Equipment (B10)

The plug load power density of 0.5 W/ft2 for "Lounge, Breakroom or Waiting Area" is not consistent with the 1.0 W/ft2 that appears in the source T24 2022 - Appendix 5.4A.

OFFM63-Equipment (B8)

The plug load power density of 12 W/ft2 for an IT room is based on "Feedback from TAG team," which seems rather subjective since this is the highest internal load of any space in the medium office building. Even if this is based on feedback from a small group, I hope that the source of the original value is something that can be cited. If it is simply a judgment call based on a small group, then gathering some sources that provide other values and show how this fits within that context would help substantiate the value.

OFFM64-Equipment (C18)

The design load for elevators of 13,930 W is based on the CEC Medium Office prototype in CBEC 2022, but it may be more appropriate to use the Title 24-2025 ACM Section 5.4.8 Elevators, Escalators and Moving Walkways, which uses 10kW per elevator. It is also not clear if the 13,930 is per elevator or total.

OFFM65-Schedules (L5:X5, L8:V8, K14:X14, K20:X20, K23:X23, K26:X26, L30:X30, L33:W33, Z33:AC33)

The Saturday occupancy schedule is not consistent with the T24 2022 - Appendix 5.4B, [Office] values, and is probably based on the other cited study: "Updated based on Plug load and schedules project with CEC, 2025," but the values for the occupancy schedule could not be found in that study, which is focused on plug loads only.

The same comment applies to the Saturday lighting schedules.

The same comment applies to the Saturday HVAC Availability schedule

The same comment applies to the Saturday infiltration schedule

The same comment applies to the Saturday heating setpoint schedule

The same comment applies to the Saturday cooling setpoint schedule

OFFM66-Schedules (L9:W9)

The lighting level on Sunday from 7 am to 6 pm is the same as it is for all other hours on Sunday, yet the building goes from no occupancy to 5% occupancy at 7 am and back down to no occupancy at 6 pm. This implies that the people who enter the office turn on no lights and that any lights left on overnight, probably mostly lighting for emergency egress, are sufficient for whatever work they are doing on Sunday. This seems unlikely.

OFFM67-Schedules (F16:AC18)

The service hot water schedule does not seem consistent with the cited source "T24 2022 - Appendix 5.4B, [Office]" for any hour. Perhaps it was copied from the "ResidentialLiving" tab.

OFFM68-Schedules (n/a)

One source that was not included in the cited report "Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual" is a PNNL study: "Energy Savings for Occupancy Based Control (OBC) of Variable Air-Volume (VAV) Systems" by J Zhang, G Liu, RG Lutes, MR Brambley from January 2013, PNNL-22072. This is a valuable source since they were specifically looking at a measure that was very sensitive to the occupancy schedule. I used it as part of the "Development of Maximum Technically Achievable Energy Targets for Commercial Buildings (Ultra-Low Energy Use Building Set) 1651-RP."

OFFM69-Schedules (n/a)

It seems unlikely that all the space in the building, other than the IT rooms, would have the same schedule. For example, the first floor lobby shows 10 people, and according to the office schedule, between 9 am and 5 pm, except for lunch time, that first floor lobby would have 95% of that value or 9.5 people. This seems very unlikely for the office building lobby, which may have a few visitors per hour passing through the lobby (for probably a few minutes each) except near opening and closing time for the building.

Another space is the breakroom on all floors and other "lobby/lounge" spaces on the middle and top floors. These are going to be occupied by the workers in the building and are most likely to be busy during lunch (rather than lower during lunch), as well as mid-morning and mid-afternoon when people typically take breaks.